

Neeloy Chakraborty

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Research Statement:

As a fifth year PhD candidate at UIUC, my research goal is to develop reliable technologies to enable safer interactions between humans and automation. To do so, I have:

- 1. studied modular approaches for **instruction-following embodied AI** robots*
- 2. developed **real-time video generation models** to allow users to remotely control robots under challenging constraints*
- 3. designed scalable algorithms to **detect anomalies in multi-agent settings***
- 4. identified **hallucinations in large language model** generations to proactively avoid failures in decision-making*

Interests: Embodied AI, Generative AI, Robot Teleoperation, Anomaly and Hallucination Detection, Reinforcement Learning

Skills: Python, PyTorch, PyTorch3D, Ollama, OpenCV, Blender, ROS, ZED Camera, Linux, Git, CARLA, Docker, C/C++

Education:

- University of Illinois Urbana-Champaign**
 - PhD in Electrical and Computer Engineering (Exp. May 2026) GPA 3.90/4.00
 - Master of Science in Electrical and Computer Engineering (2021 – 2023) [Thesis] GPA 3.83/4.00
 - Bachelor of Science in Computer Engineering (2017 – 2021) [Thesis] GPA 3.75/4.00

Current Research Projects:

- Creative 3D Asset Generation** Summer 2025 –
Advisor: Dr. Guan-Ming Su
Developing 3D asset generation models conditioned on different modalities
Dolby Laboratories
- Hallucination Detection and Mitigation for Foundation Models in Decision-Making Tasks** Spring 2024 –
Advisors: Professor Melkior Ornik and Professor Katherine Driggs-Campbell
Formulating pipelines to identify hallucinations in large (visual) language models
University of Illinois

Past Research Projects:

- Image Generation for Delay-Compensated Video Feeds in Mobile Robot Teleoperation** Summer 2023 – Fall 2024
Advisor: Professor Katherine Driggs-Campbell
Developing real-time image generation methods combining depth foundation models and neural rendering
University of Illinois
- Co-operative Traffic Congestion Mitigation** Spring 2022 – Spring 2024
Advisors: Professor Cathy Wu and Professor Katherine Driggs-Campbell
Evaluating human-in-the-loop traffic congestion mitigation policies with real users in a car simulator
University of Illinois
- Highway Vehicle Anomaly Detection** Fall 2021 – Spring 2024
Advisor: Professor Katherine Driggs-Campbell
Designing frameworks for identifying anomalies on roads from multi-modal sensor data
University of Illinois
- Modular Embodied Agent for Alexa Arena** Fall 2022 – Spring 2023
Advisor: Professor Julia Hockenmaier
Leading development of an instruction-following embodied agent with action prediction and computer vision
University of Illinois

- **Decentralized Robot Crowd Navigation** Fall 2019 – Spring 2023
Advisor: Professor Katherine Driggs-Campbell University of Illinois
Researching a novel network to guide a robot to reach a goal state while avoiding colliding with other agents
 - **Koopman Models for Reinforcement Learning** Summer 2022
Advisors: Dr. Kaushik Balakrishnan and Dr. Devesh Upadhyay Ford Motor Company
Exploring benefits of combining Koopman theory with model-based RL for driving tasks
 - **Hierarchical Self-Imitation Reinforcement Learning with Sparse Rewards** Fall 2020 – Spring 2021
Advisor: Professor Katherine Driggs-Campbell University of Illinois
Applying hierarchical and self-imitation learning to long horizon single-agent environments with sparse rewards
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Publications

(* and + denote equal contribution and noteworthy comments are in red)

Pre-prints:

- **N. Chakraborty**, J. Pohovey, M. Ornik, and K. Driggs-Campbell. “Adaptive Stress Testing Black-Box LLM Planners,” Under review 2025 [[Paper](#)]

Journal Articles:

- S. Liu, H. Xia, F. C. Pouria, K. Hong, **N. Chakraborty**, Z. Hu, J. Biswas, and K. Driggs-Campbell. “HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments,” IEEE Transactions on Automation Science and Engineering (T-ASE), 2025 [[Paper](#)] [[Website](#)] [[Code](#)]
- A. Hasan, **N. Chakraborty**, H. Chen, C. Wu, and K. Driggs-Campbell. “Lessons in Cooperation: Driver Sentiments towards Real-Time Advisory Systems,” IEEE Intelligent Transportation Systems Magazine (ITSM), 2025 [[Paper](#)]
- **N. Chakraborty**, M. Ornik, and K. Driggs-Campbell. “Hallucination Detection in Foundation Models for Decision-Making: A Flexible Definition and Review of the State of the Art,” ACM Computing Surveys (CSUR), 2025 [[Paper](#)]
- A. Hasan, **N. Chakraborty**, H. Chen, J-H. Cho, C. Wu, and K. Driggs-Campbell. “Cooperative Advisory Residual Policies for Congestion Mitigation,” ACM Journal on Autonomous Transportation Systems (JATS), 2024 [[Paper](#)]
- T. Ji*, **N. Chakraborty***, A. Schreiber, and K. Driggs-Campbell. “An Expert Ensemble for Detecting Anomalous Scenes, Interactions, and Behaviors in Autonomous Driving,” International Journal of Robotics Research (IJRR), 2024 [[Paper](#)]

Conference Articles:

- **N. Chakraborty***, Y. Fang*, A. Schreiber, T. Ji, Z. Huang, A. Mihigo, C. Wall, A. Almana, and K. Driggs-Campbell. “Towards Real-Time Generation of Delay-Compensated Video Feeds for Outdoor Mobile Robot Teleoperation,” IEEE International Conference on Robotics and Automation (ICRA) 2025 [[Paper](#)] [[Website](#)] [[Code](#)] [[Dataset](#)] (Leading cross-departmental team of researchers with diverse skillsets)
- P. Chang, S. Liu, T. Ji, **N. Chakraborty**, K. Hong, and K. Driggs-Campbell. “A Data-Efficient Visual-Audio Representation with Intuitive Fine-tuning for Voice-Controlled Robots,” Conference on Robot Learning (CoRL) 2023 [[Paper](#)] [[Website](#)]
- A. Hasan, **N. Chakraborty***, H. Chen*, J-H. Cho, C. Wu, and K. Driggs-Campbell. “PeRP: Personalized Residual Policies for Congestion Mitigation Through Co-operative Advisory Systems,” IEEE International Conference on Intelligent Transportation Systems (ITSC) 2023 [[Paper](#)] [[Website](#)] [[Code](#)]
- **N. Chakraborty***, R. Sidhu*, B. Abdullai*, H. Chen*, N. Ravi*, A. Ankur, D. Prasad, and J. Hockenmaier. “BEAST: Building an Embodied Action-prediction System with Trajectory data,” Alexa Prize SimBot Challenge Proceedings 2023 [[Paper](#)] [[Website](#)] (University team leader and semi-finalist in inaugural Amazon SimBot Challenge)

- S. Liu, P. Chang, Z. Huang, **N. Chakraborty**, K. Hong, W. Liang, D. L. McPherson, J. Geng, and K. Driggs-Campbell. "Intention Aware Robot Crowd Navigation with Attention-Based Interaction Graph," IEEE International Conference on Robotics and Automation (ICRA) 2023
[Paper] [Website] [Code] **(Best poster award at IROS 2023 Last-Mile Robotics Workshop)**
- **N. Chakraborty**, A. Hasan*, S. Liu*, T. Ji*, W. Liang, D. L. McPherson, and K. Driggs-Campbell. "Structural Attention-based Recurrent Variational Autoencoder for Highway Vehicle Anomaly Detection," IFAAMAS International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023
[Paper] [Website] [Code] **(Accepted as full paper with 23.3% acceptance rate and received student scholarship)**
- S. Liu, P. Chang, H. Chen, **N. Chakraborty**, and K. Driggs-Campbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference," IEEE International Conference on Robotics and Automation (ICRA) 2022
[Paper] [Website] [Code]
- S. Liu*, P. Chang*, W. Liang+, **N. Chakraborty+**, and K. Driggs-Campbell. "Decentralized Structural-RNN for Robot Crowd Navigation with Deep Reinforcement Learning," IEEE International Conference on Robotics and Automation (ICRA) 2021 [Paper] [Website] [Code]

Workshop Papers:

- A. Hasan, **N. Chakraborty**, C. Wu, and K. Driggs-Campbell. "Towards Co-operative Congestion Mitigation," Shared Autonomy in Physical Human-Robot Interaction: Adaptability and Trust, ICRA 2022 Workshop [Paper]

Theses:

- "Detecting the Unpredictable: Advanced Trajectory and Camera-Based Anomaly Detection Methods for Vehicles," Illinois Digital Environment for Access to Learning and Scholarship (IDEALS), 2023 [Paper]
- "Hierarchical Self-Imitation Learning in Single-Agent Sparse Reward Environments," Illinois Digital Environment for Access to Learning and Scholarship (IDEALS), 2021 [Paper]

Patents Applied:

- K. Balakrishnan, **N. Chakraborty**, and D. Upadhyay. "Model-based Reinforcement Learning," US20240320505A1 [Paper]

Industry Experience:

- **PhD Research Intern in Sight Lab at Dolby Laboratories** Summer 2025
Researching greenfield project within the space of creative 3D asset generation
- **Research and Advanced Engineering Intern in Core AI/ML at Ford Motor Company** Summer 2022
Designing sample efficient model-free + model-based RL methods
Outperforming classical PID controllers by 41% in complex autonomous vehicle use case
Conducting both independent and collaborative research and guiding experimental design according to results
- **Perception Engineering Intern in Autonomy Team at Brunswick i-Jet Lab** Summer 2021
Localizing swimmers around boats using filtering, tracking, and computer vision techniques
Researching sensors and communicating with sensor companies to increase autonomy stack capabilities
Analyzing functional safety standards practiced at company and presenting findings to global management team
- **Engineering Intern in Global CAD Team at Qualcomm** Summer 2020
Building generalized data gathering solutions to support Design For Test pipeline [Website]
Leading design process of a base framework for data gathering tool
Collaborating and adapting with international teams to consider multiple perspectives

- Global Management Trainee Intern in Solutions at Anheuser Busch** Summer 2019
 Identifying root causes of a multi-million-dollar annual problem with Six Sigma LEAN exercises [\[Website\]](#)
 Implementing short- and long-term process solutions leveraging technology with an annual ROI of \$1.5M
 Pitching solutions to multidisciplinary teams in the People department including managers and directors
- Global Management Trainee Intern in Logistics at Anheuser Busch** Summer 2018
 Increasing productivity of critical decision-making team by developing clear visualizations [\[Website\]](#)
 Creating effective data visualizations with Qlik Sense and SQL databases
 Connecting with multidisciplinary logistics teams and interns

Teaching Experience:

- Lab Teaching Assistant for Introduction to Robotics (ECE 470)** Fall 2021 – Fall 2022
 Guiding students to program a UR3 arm with ROS and implement kinematics and computer vision
- Undergraduate Course Assistant for Digital Systems Laboratory (ECE 385)** Fall 2019 – Spring 2021
 Providing impactful assistance to students on TTL and System Verilog hardware labs

Coursework:

- Artificial Intelligence**
 - Intro to Artificial Intelligence** search, classification, NLP, CV, robotics
 - Intro to Deep Learning** linear classifiers, MLPs, CNNs, RNNs, generative models, deep RL
 - Intro to Reinforcement Learning** RL foundations, model-free, policy gradient, exploration/exploitation
 - Pattern Recognition** nearest neighbor, regression, optimization primal/dual, SVM, learning theory
 - Generative AI Models** normalizing flows, VAEs, GANs, RNNs, transformers, applications, explainability
 - Random Processes** measure theory, convergence of sequences, estimators, random walks, martingales
 - Computer Vision** image processing, fitting, geometric vision, recognition
 - Statistical Learning Theory** concentration inequalities, empirical risk minimization, generalization bounds
- Robotics**
 - Intro to Robotics** rigid motion, forward/inverse kinematics, motion planning, vision
 - Human-Centered Robotics** design robots that interact with people safely
 - Principles of Safe Autonomy** path planning, localization, lane detection, safety verification
 - Control Systems** dynamic models and response, root locus/frequency response techniques
 - Control System Theory & Design** state space models, stability, controllability, observability, tracking
 - Hybrid Systems and Control** stability, observability, etc. of switched and hybrid control systems
- Hardware/Software Systems**
 - Digital Systems Lab** logic types, storage, design tradeoffs, FPGAs, microprocessor design
 - Computer Systems Engineering** operating system design, I/O, synchronization, interrupts, virtualization

Additional Projects:

- Efficient FPGA Smart Home Security Camera System (Project Watchdog)** Apr 2019 – Oct 2019
 Leading hardware design of accelerated IoT security system
 Student Group Project
 Regional Finalist in InnovateFPGA 2019 [\[Website\]](#)
- Brain Computer Interface Platform for IoT Applications (Project HackMe)** Feb 2019
 Leading data analytics and storage team
 Student Group Project
 HackIllinois 2019 runners-up and sponsor award winners [\[Website\]](#)
- Human interactive Balancing Security Robot** Aug 2016 – Jun 2017
 Advisor: Dr. George Kantor
 Carnegie Mellon University
 Engineering a human interactive segway security robot to roam halls of an institution for safety [\[Website\]](#)

- Robotics Project at ZeGoBeast LLC Pittsburgh**
May 2016 – Jun 2017
 Advisors: Mr. Daniel Goncharov and Mr. Alex Thomson
 ZeGoBeast LLC Pittsburgh
 Building & improving the wooden ZeGoBeast Electric robot and presenting at New York Maker Faire
 [[Website](#)]

Organizations & Extracurriculars:

- Eta Kappa Nu (HKN) Electrical Engineering Honor Society**
Aug 2019 –
 Dedicated to serving the ECE & Engineering student body by providing services to help students succeed
 Member
 Holding course review sessions and sharing university experiences with other students
- Human-Centered Autonomy Lab**
Aug 2021 – Aug 2024
 Connecting passionate undergraduate researchers with graduate mentors
 Research Volunteer Coordinator
- iRobotics Combotics & Projects Student Organization**
Aug 2017 – May 2018
 Collaborated on the mechanical design team by developing CAD designs that model the real robot
 Member
- Children's Library of Pittsburgh**
Jun 2012 – Jun 2017
 Shelves, counted, & organized books in the Children's section of the Main Library
 Volunteer
 Supported in the planning & development of tech programs to introduce children to programming

Academic Honors and Services

Awards:

- IROS Last-Mile Robotics Workshop Best Poster 2023
- AAMAS Student Scholarship 2023
- Graduated BS with Honors 2021
- Eta Kappa Nu Member 2019 – present
- AB InBev GMT Full Tuition Scholarship 2017 – 2021
- James Scholar 2017 – 2021
- Dean's List 2019 – 2021
- InnovateFPGA Regional Finalist 2019
- HackIllinois 2019
 - 2nd Place
 - Caterpillar Sponsor Award
 - Particle Sponsor Award

Students Mentored:

- Abdulrahman Almana 2024; KAUST Gifted Student Program at UIUC
- Cassidy Wall 2024; BS Systems Engineering and Design
- Yixiao Fang 2023 – 2024; BS Computer Engineering 2023; Now studying MS at UIUC
- Chun-Yen Yeh 2024; MEng Autonomy & Robotics
- ChuLin Huang 2024; MEng Autonomy & Robotics
- Louis Sungwoo Cho 2023; BS Civil and Environmental Engineering 2023; Now studying MS at UIUC
- Wenjian Zhao 2023; BS Engineering Mechanics 2023; Now studying MEng at MIT
- Devika Prasad 2022 – 2023; BS Computer Science, Now at Applied Intuition
- Haomiao Chen 2022 – 2023; BS Physics; Now studying PhD at Cornell
- Blerim Abdullai 2022 – 2023; BS Electrical and Computer Engineering 2023; Now studying MS at U of T
- Nikil Ravi 2022 – 2023; BS Computer Science 2023; Now studying MS at Stanford

Journal and Conference Reviews:

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| • IEEE International Conference on Robotics and Automation (ICRA) | 2022, 2023, 2025, 2026 |
| • Conference on Robot Learning | 2025 |
| • IEEE International Conference on Intelligent Robots and Systems (IROS) | 2025 |
| • IEEE MultiMedia | 2025 |
| • IEEE Robotics and Automation Letters (RAL) | 2023, 2024 |
| • IEEE Transactions on Control Systems Technology (TCST) | 2024 |
| • Springer International Journal of Social Robotics | 2024 |
| • IEEE Transactions on Network Science and Engineering (TNSE) | 2023 |